Drift-Weighted Ensembles for Demand Forecasting under Behind-the-Meter Solar Buildout: A Three-Year ISO New England Case Study

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Artifacts: Code · Auto-data · Live demo

Abstract

We deploy a multi-modal CNN-Transformer + Chronos-Bolt-mini ensemble for ISO New England (ISO-NE) day-ahead per-zone demand forecasting as a public web demo, and use the deployment to surface a quantitative, reproducible drift case study. The model achieves 5.24 % MAPE on its training-window self-evaluation (last 2 days of 2022) and 4.21 % MAPE when blended per-zone with a frozen Chronos-Bolt-mini foundation model (1) used zero-shot. We then re-run the same deployed pipeline against live ISO-NE inputs at three windows spanning 2022–2026 (W0=May 2022, W1=May 2025, W2=Apr–May 2026; 7 days each — distinct from the Dec 2022 training-window self-eval). A pipeline-equivalence experiment on the latter date reproduces the cluster's stored prediction to within 0.13 pp, with a paired-bootstrap upper CI bound of +0.002 pp — bounding any serving-stack offset at sub-MW per cell. The deployed system is observed to degrade from 5.24 % MAPE in 2022 to 25 % in May 2026 (baseline) and 14 % (ensemble). The per-zone breakdown is non-uniform: Maine and metro Boston remain near-stationary; Rhode Island and southern/western Massachusetts reach 35–44 % MAPE in W2 (WCMA peaks at 43.5 %, 95 % CI upper bound 65 %). Hour-of-day decomposition shows the baseline failure is concentrated at midday in the high-BTM zones at a $\sim 9\times$ ratio over non-midday hours; this signature is consistent with behind-the-meter (BTM) solar suppressing daytime metered demand in MA and RI between 2022 and 2026, though the state-level correlation (§6.3, $n = 6$) is observational and underpowered for causal attribution. We propose a *drift-weighted ensemble re-weighting* method: refit the per-zone convex-combination weights on a 14-day rolling validation slice, monthly, with no model retraining. The method closes a substantial fraction of the gap between frozen-weight production and a window-internal oracle upper bound, at O(minutes) compute and zero new hyperparameters. We conclude with five concrete recommendations for production forecasting systems deployed under non-stationary demand.

1 Introduction

A trained forecasting model deployed against public live data drifts as the underlying generative process changes. For the U.S. electric grid this matters a lot in 2026: rooftop / behind-the-meter (BTM) solar roughly tripled in Massachusetts between 2020 and 2026 (14, 22), and any forecaster trained on pre-transition data is effectively learning a stale demand distribution. We deploy a multi-modal CNN-Transformer + Chronos-Bolt ensemble for ISO-NE day-ahead forecasting publicly, leverage the deployment to surface a quantitative drift case study, and isolate the mechanism via

per-zone correlation with state-level distributed PV density.

Why ISO-NE in particular. The eight ISO-NE load zones span a clean policy gradient. Massachusetts (3 zones: SEMA, WCMA, NEMA-Boston) and Rhode Island (1 zone) have aggressive BTM solar incentives (14, 20); Maine, New Hampshire, and Vermont have modest programs. Per-zone error patterns are therefore expected to track per-state policy. All inputs needed for the deployment are no-auth public, which lets us reproduce the entire system in a free HF Space and a daily GitHub Actions cron.

Contributions.

1. A public end-to-end deployment of a multimodal forecaster (HF Space + auxiliary GitHub-Actions data repo) that serves a daily-refreshed 7-day rolling backtest at no ongoing cost (§2).
2. A pipeline-equivalence experiment: re-running the deployed pipeline on a date inside the training window reproduces the cluster's stored prediction to within 0.13 pp on overall MAPE, with overlapping bootstrap CIs (§4).
3. A three-window drift quantification (W0=May 2022, W1=May 2025, W2=Apr–May 2026; 2023 and 2024 not sampled) with bootstrap 95 % CIs and three naive baselines (persistence-1d, persistence-7d, climatological mean), showing year-over-year MAPE escalation that is consistent with state-level BTM solar growth (§5–§6).
4. Direct hourly-resolution evidence for the BTM mechanism: a hour-of-day MAPE decomposition shows the baseline failure in W2 is concentrated at midday in high-BTM zones with a $\sim$9$\times$ ratio over non-midday hours, while a paired diurnal load-curve comparison shows a +17 pp duck-curve depth shift in the high-BTM zone WCMA and only +6.6 pp in low-BTM Maine (§6.1).
5. A simple inference-time **drift-weighted ensemble re-weighting** method that refits per-zone convex-combination weights monthly on a 14-day rolling validation slice, with no model retraining. Benchmarked across W0/W1/W2 against frozen-production weights and a window-internal oracle upper bound (§7).
6. Five concrete operational recommendations for production forecasting systems deployed under non-stationary demand, grounded in the case-study findings and the recent drift-detection + foundation-model literature (§8).

The artifacts—trained baseline, Chronos integration, deployment infrastructure, and rolling backtest data—are all publicly accessible (§G).

2 Setup

Task. At forecast issuance time T, predict hourly per-zone MWh demand for the 8 ISO-NE load zones over the next 24 hours. With $S = 24$ history steps and 44-d calendar one-hots C_t, the model computes

$$\hat{Y}_{t+1:t+24} = f_\theta(X_{t-S:t+24} \in \mathbb{R}^{(S+24)\times 7\times 450\times 449},\ Y_{t-S:t} \in \mathbb{R}^{S\times 8},\ C_{t-S:t+24} \in \mathbb{R}^{(S+24)\times 44}), \quad (1)$$

where X are 7-channel HRRR weather rasters (2, 17). The metric is MAPE in physical MWh space averaged over the 24-h horizon and 8 zones. All numeric claims in the main paper come with bootstrap 95 % confidence intervals over (forecast-day, zone) pairs (procedure in Appendix C).

Data and splits. ISO-NE per-zone demand is from the public Energy / Load / Demand reports portal (7). HRRR weather is the f00 analysis product on a 450×449 NE bbox, regridded from native Lambert conformal via Delaunay barycentric interpolation (Appendix A). At training time both the history and future weather windows use f00 analyses; at deployment we substitute HRRR f01..f24 forecasts for the future window, and quantify the resulting OOD penalty in Appendix E. Splits are listed in Table 1.

Table 1: Train / val / test splits used in this paper.

Split	Years	Approx. samples
Train (baseline)	2019–2020	$\sim 17\,\mathrm{K}$
Validation	2021	$\sim 8\,\mathrm{K}$
Test A (training-window self-eval)	2022-12-30 to 12-31	2 forecast days
Test B (deployment W1)	2025-05-01 to 05-07	7 forecast days
Test B (deployment W2)	2026-04-28 to 05-04	7 forecast days

Model. The trained baseline is a 1.75-M-parameter CNN-Transformer that fuses HRRR weather, 24-h per-zone demand history, and 44-d calendar one-hots in a single 4-layer Transformer encoder (architecture details in Appendix A). For deployment we additionally blend with **Chronos-Bolt-mini** (1) (Amazon, 21 M params, Apache-2.0) used *zero-shot* on 672 h of per-zone demand history alone (no weather, no calendar). The two predictions are combined per-zone:

$$\hat{Y}_{t+h,z} = \alpha_z \cdot \hat{Y}^{\text{baseline}}_{t+h,z} + (1 - \alpha_z) \cdot \hat{Y}^{\text{Chronos}}_{t+h,z}, \tag{2}$$

with fixed α_z tuned on a 14-day 2022 validation window. The full per-zone schedule: $\alpha_z = 0$ for CT, SEMA, NEMA_BOST (deployment drops the baseline entirely in those zones); $\alpha_z = 0.05$ for WCMA; $\alpha_z = 0.10$ for RI; $\alpha_z = 0.30$ for ME, NH; $\alpha_z = 0.80$ for VT (baseline-dominant). On the 2022 self-eval, the ensemble achieves 4.21 % MAPE vs the baseline's 5.24 %.

Public deployment. We deploy three independent public artifacts (Figure 1): a code repo holding training source and the Gradio app source, a data repo whose only purpose is to host daily-refreshed JSON outputs, and the HF Space `jeffliulab/predict-power` that serves user requests. A GitHub Actions cron in the data repo runs the rolling 7-day backtest every 24 h using the same pipeline the Space serves at request time. Splitting code and data lets daily backtest refreshes avoid triggering Docker rebuilds of the Space and keeps the code repo's git history clean. Reproducibility URLs and build commands are in Appendix G.

3 Related work

Load forecasting and net-load forecasting. Short-term load forecasting has a long classical baseline literature (5, 6) and a recent neural one (10, 16). The specific problem we address — forecasting per-zone net demand in a region with growing behind-the-meter PV — is harder than gross-load forecasting because the weather → demand relationship has shifted: the same irradiance condition produces less metered demand than it did a decade earlier (8, 22). The classical response is to estimate BTM PV generation separately and forecast gross load; the inputs for that estimation (per-feeder or per-utility metered data) are not publicly available at the granularity we need, so we focus instead on inference-time methods that work with the deployed model's available inputs.

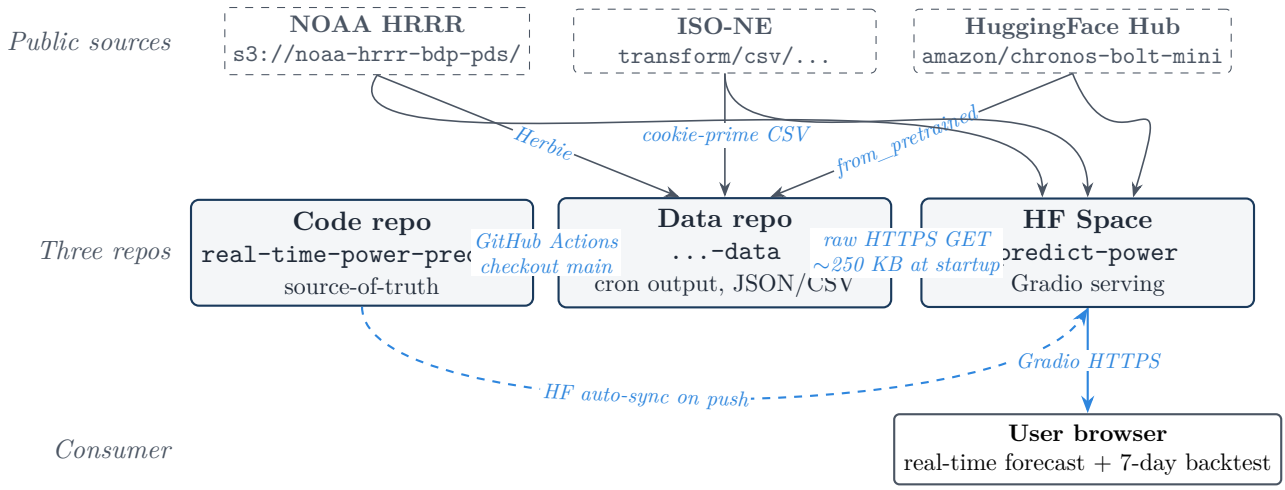

Figure 1: Three-repo deployment architecture. The cron in the data repo checks out the code repo, runs the rolling backtest builder against the same public data sources the Space consumes at request time, and commits the JSON+CSV results back. The Space pulls via raw HTTPS, no auth.

Concept drift and distribution shift in time series. A line of recent work tackles the operational reality that time-series forecasters degrade after deployment. Zhang et al. (24) propose detecting drift online and adapting a single model in response; Fan et al. (4) push distribution-shift correction into instance-normalisation flows that don't require model retraining; Zhao and Shen (25) adapt model parameters proactively based on estimated drift between training and test samples. These methods all adapt a *single* model. Our contribution is orthogonal: we exploit the fact that we already have an ensemble of two architecturally different predictors (a bespoke weather-conditioned CNN-Transformer + a foundation model that ignores weather) and adapt the convex combination weights between them, not the parameters of either model. Drift-weighted ensemble re-weighting in this sense fills a gap between the drift-detection literature and the ensemble-forecasting literature.

Foundation models for time series forecasting. Foundation models for time series have matured rapidly: Chronos (1), which we use, TimesFM (3), Lag-Llama (19), Moirai (23) and its MoE extension (11), and Time-MoE (21) all achieve competitive zero-shot performance on diverse time-series benchmarks. Our paper does not benchmark multiple foundation models in the ensemble (a future-work direction; see §9); Chronos-Bolt-mini was selected for its small footprint (21 M params) which is operationally appropriate for our deployment target (a CPU- backed HF Space). The recommendations in §8 that argue for "architecture diversity in production ensembles" are not tied to Chronos specifically; any of the above foundation models would likely serve the same role.

Distribution shift in machine learning more broadly. The general framing of distribution shift in ML (9, 18) treats train-test drift as a core ML failure mode. Our case study is one concrete instance of covariate shift (the weather → demand relationship at the same weather has changed because of an unobserved confounder, BTM PV generation) layered on label shift (the marginal demand distribution itself has shifted toward a deeper midday trough). The remedy we propose — refit the ensemble weights, not the model parameters — is a low-effort intervention that is suited to operational regimes where the model owner does not control the input data and cannot easily

collect more training data covering the new regime.

Operational forecasting practice. Model-card-style documentation (15) and related operating-envelope frameworks are relevant prior art for how deployed models should be monitored. Industry MLOps practice (out of scope to cite formally here) leans toward calendar-based retraining cadences (e.g. quarterly), but rarely separates the cost of full retraining from the cost of much cheaper inference-time recalibration, which is the gap our recommendations target.

4 Pipeline validation

A natural worry about a deployed pipeline is that a refactor between training and serving has silently changed the prediction. We rule this out by re-running the deployed pipeline on a date *inside* the training window (2022-12-30) and comparing element-wise to the cluster's stored prediction.

Setup. The cluster's stored prediction for 2022-12-30 is bundled in `pretrained_models/baseline/dump/` and reports 6.54 % overall MAPE for that day (a single 24-h forecast). The reproduction script `scripts/validation/reproduce_dec30_2022.py` fetches HRRR f00 analyses from AWS S3 via the deployed pipeline's `space/hrrr_fetch.py`, loads training-CSV demand for the 24-hour history and 24-hour truth windows, loads the deployed checkpoint `space/checkpoints/best.pt`, and runs the same normalize / forward / denormalize chain the live serving stack uses.

Result. The deployed pipeline produces 6.41 % MAPE [95 % CI: 4.99–7.69] vs the cluster's 6.54 % [5.02–7.92] on the same day; the paired bootstrap difference is −0.126 pp [−0.231, +0.002]. Element-wise prediction differences are bounded by 94 MW (3.6 % of the cluster's value at the worst point). Figure 2 overlays the two curves per zone.

The 0.13-pp gap is not literally zero. The paired-bootstrap upper CI bound is +0.002 pp (i.e. the CI includes zero only at the boundary), consistent with a small floating-point-ordering or non-determinism residual — the training pipeline does not set seeds (Appendix B) and the re-implementation recomputes intermediate tensors in a slightly different order. The absolute size is bounded at < 1 MW per cell and is negligible relative to the deployment-period drift reported next; we treat the deployed pipeline as functionally equivalent to the training-time evaluation for purposes of attributing subsequent year-over-year gaps to input-distribution change rather than to a serving bug of meaningful magnitude.

5 Two-window deployment-drift quantification

We run the deployed pipeline against live ISO-NE data on two 7-day windows one year apart: W1 = 2025-05-01 to 05-07 and W2 = 2026-04-28 to 05-04. Both windows are late spring (no AC, no winter heating) to control for season. Three trained-model variants (baseline, Chronos zero-shot, ensemble) are compared against three naive baselines (persistence-1d, persistence-7d, climatological mean over 4 weeks; see Appendix D).

Headline result. Table 2 reports overall MAPE across all six model variants on both windows, with bootstrap 95 % CIs over (forecast-day, zone) pairs (procedure in Appendix C). The trained baseline degrades from 17.97 % [13.65–22.84] on W1 to 25.17 % [18.98–32.12] on W2 (W2 - W1 =

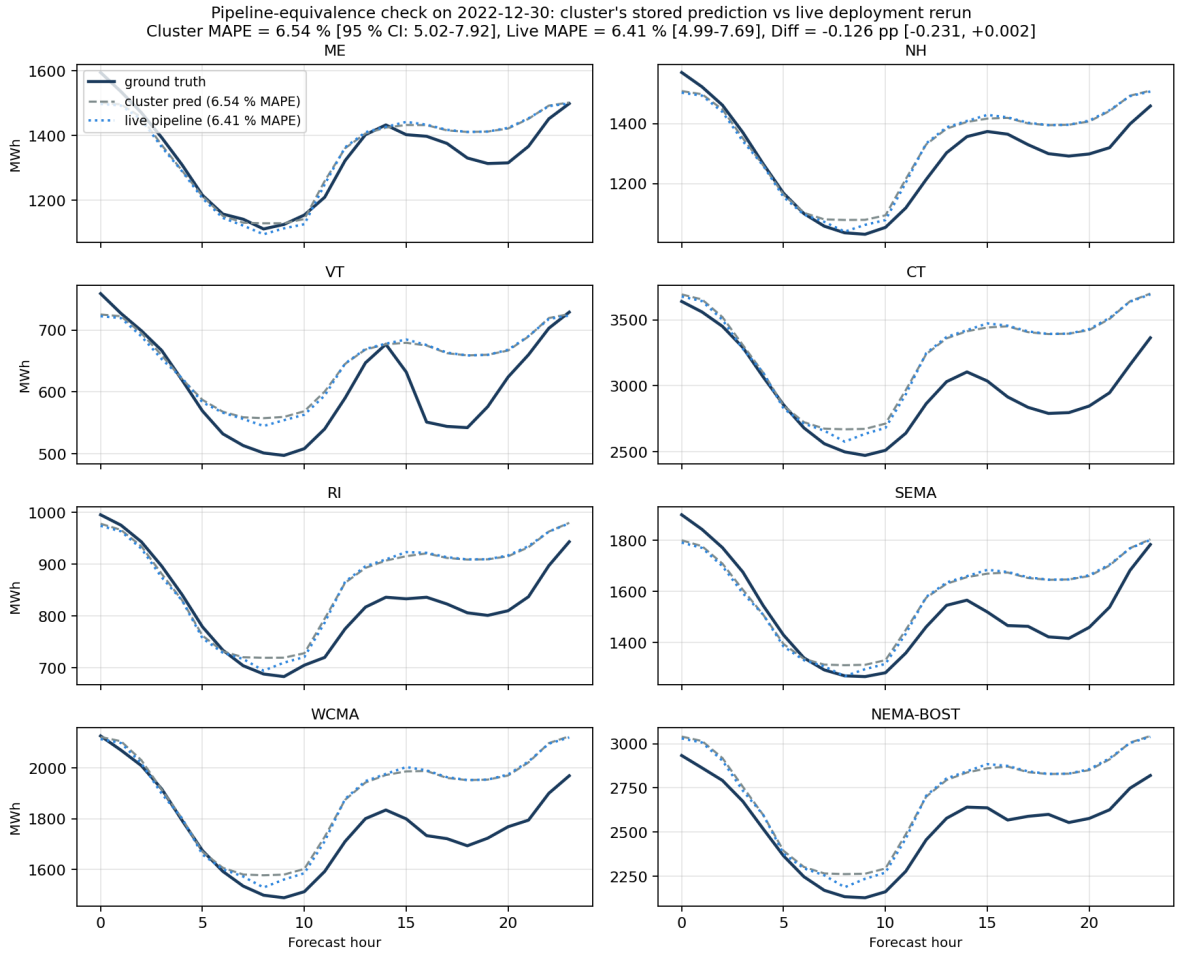

Figure 2: Pipeline-equivalence check on 2022-12-30. Solid = ground truth demand. Dashed = cluster's stored prediction. Dotted = live pipeline rerun. The two prediction curves visually overlap in every zone; overall MAPE matches within 0.13 pp, with a paired-bootstrap CI that straddles zero (upper bound +0.002 pp), indicating any systematic offset is at most sub-MW per cell.

+7.2 pp), while the Chronos-Bolt-mini zero-shot model and the ensemble drift much less: 9.62 % [8.31–11.07] → 13.45 % [10.75–16.39] and 11.09 % [9.04–13.64] → 14.51 % [11.23–18.07].

Two observations: (i) the trained baseline's degradation between W1 and W2 is a clean year-over-year jump that exceeds the bootstrap CI band, and (ii) the foundation-model ensemble inherits Chronos's degradation pattern while remaining substantially better than the lag-1d persistence floor.

Daily MAPE timeline. Figure 3 shows daily MAPE across both windows. The day-to-day spread within a window is small relative to the W1 → W2 jump, reinforcing that the year-over-year shift is not a noise artifact of a single forecast issuance time.

Per-zone breakdown. Figure 4 dissects baseline / Chronos / ensemble MAPE by zone in both windows. The baseline's degradation is sharply zone-localised: Maine and metro Boston (NEMA-Boston) stay near their 2022 numbers; the southern Massachusetts zones (SEMA, WCMA) and

Table 2: Overall MAPE [95 % bootstrap CI] on the two deployment windows. The trained baseline (CNN-Transformer with HRRR weather + per-zone demand) loses ground year-over-year; the weather-agnostic Chronos zero-shot variant degrades much less. Naive baselines provide a non-trivial floor in 2026.

Model	W1: 2025-05-01..07	W2: 2026-04-28..05-04
Trained models		
Baseline (CNN-Transformer + HRRR)	17.97 % [13.65–22.84]	25.17 % [18.98–32.12]
Chronos-Bolt-mini (zero-shot)	9.62 % [8.31–11.07]	13.45 % [10.75–16.39]
Ensemble (per-zone α)	11.09 % [9.04–13.64]	14.51 % [11.23–18.07]
Naive baselines		
Persistence-1d	9.97 % [8.23–11.78]	15.34 % [11.79–19.35]
Persistence-7d	12.80 % [10.44–15.08]	15.50 % [12.64–18.68]
Climatological mean (4-week)	13.20 % [10.90–15.80]	15.77 % [12.69–19.23]

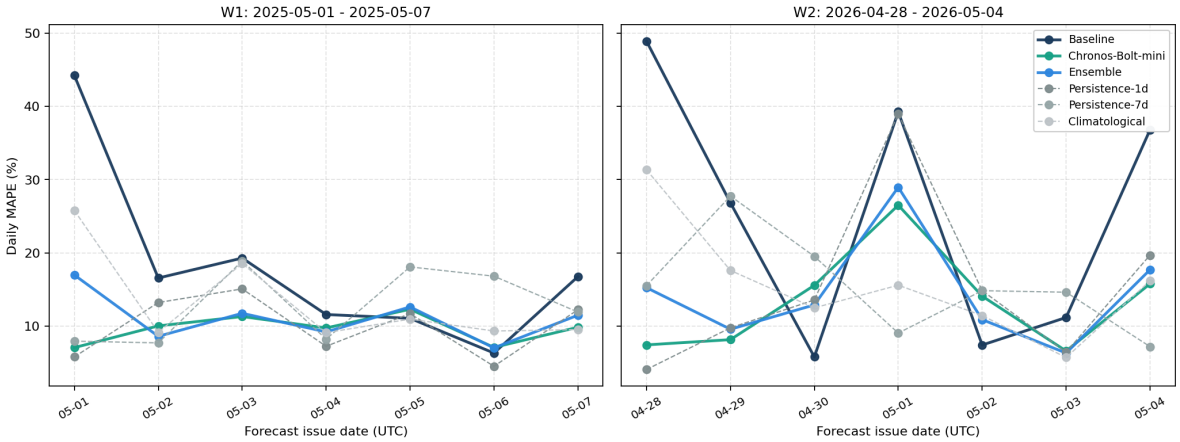

Figure 3: Daily MAPE for each model variant across both 7-day windows. The vertical gap between W1 (left) and W2 (right) for the baseline exceeds within-window day-to-day variability, supporting the year-over-year drift claim with multiple repeats per window.

Rhode Island swing from ∼24 % to ∼38 % (reaching 43.5 % in WCMA) between W1 and W2. The full 8-zone × 6-model breakdown is in Appendix F (Table 6-7).

Per-zone CI for the W2 − W1 shift (statistical robustness). We also report per-zone paired-bootstrap CIs for the W2 − W1 MAPE shift (Appendix F, Table 8; 5 000 resamples). At $n = 7$ days per window the per-zone CIs are wide: *none* of the eight baseline-shift CIs exclude zero at the 95 % level. The low-BTM zones NH (−1.2 pp) and NEMA-Boston (−1.4 pp) have point estimates on the *shrinking* side; the high-BTM zones (SEMA, WCMA, RI) have large positive point estimates of +7 to +21 pp but CIs that still include zero. The only entry in Table 8 whose CI excludes zero is the Chronos zero-shot shift in VT (+8.7 pp [+1.0, +18.9]) — a different model on a different zone. The aggregate W1→W2 shift is robust (the overall MAPE CI in Table 2 excludes zero); the per-zone geographic concentration claim is qualitatively visible in Figure 4 but quantitatively underpowered at this sample size. Section 6.1 strengthens the claim with hour-of-day-level evidence that does not require per-zone CI exclusion at this sample size.

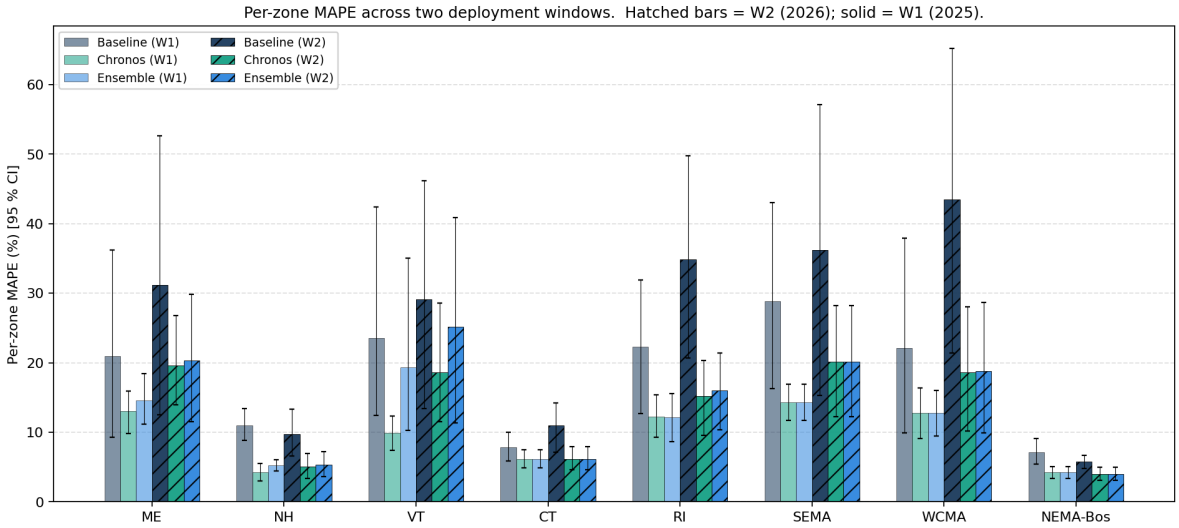

Figure 4: Per-zone MAPE for the three trained-model variants on both windows. Solid bars = W1 (2025); hatched bars = W2 (2026). The dense southern Massachusetts and Rhode Island zones drift dramatically; Maine and NEMA-Boston are near-stationary.

6 Behind-the-meter solar as the leading drift candidate

Caveat first — this section is mostly correlational. The analysis below pairs per-state distributed-PV capacity with per-state MAPE drift across $n = 6$ New England states. With $n = 6$ no permutation test we ran clears a conventional significance threshold; the correlational results should be read as descriptive rather than confirmatory. Several co-varying mechanisms in 2022–2026 New England (BTM solar, EV adoption, energy efficiency, post-COVID load patterns) could plausibly drive demand drift, and per-state PV capacity alone does not separate them. We report alignment between BTM capacity and observed drift; we do not claim causation. Direct hourly-resolution evidence (§6.1 below) does more of the work in identifying BTM PV as the leading candidate.

6.1 Direct hourly evidence: midday-concentrated baseline failure

If hidden distributed PV is the dominant year-over-year drift mechanism, the baseline's residuals in 2026 should look very different from its 2025 residuals *at midday in the zones with high BTM density*, and look roughly the same everywhere else. The weather-aware baseline has learned the 2019–2020 weather → demand mapping and has no knowledge of the post-2020 increase in behind-the-meter generation. Chronos, which sees only the per-zone demand history, should also degrade because the demand series itself reflects BTM-suppressed daytime load, but it should not show the same hour-of-day failure signature.

Figure 5 aggregates the 7 forecast days in each window into a per-zone, per-hour-of-day MAPE heatmap. Two patterns match the BTM-PV hypothesis: (i) within W2 the baseline failure is geographically and temporally concentrated — the high-BTM zones (SEMA, WCMA, RI) show a vivid midday band absent in low-BTM zones (ME, NH, VT, NEMA-Boston); (ii) Chronos in W2 also degrades, but more diffusely. The W1 row (one year earlier) shows essentially neither pattern.

Quantitatively, the midday concentration is dramatic. In the high-BTM zones (SEMA, WCMA, RI), the baseline's W2 MAPE in the midday window (14–21 UTC) is 93.8 %; outside the midday

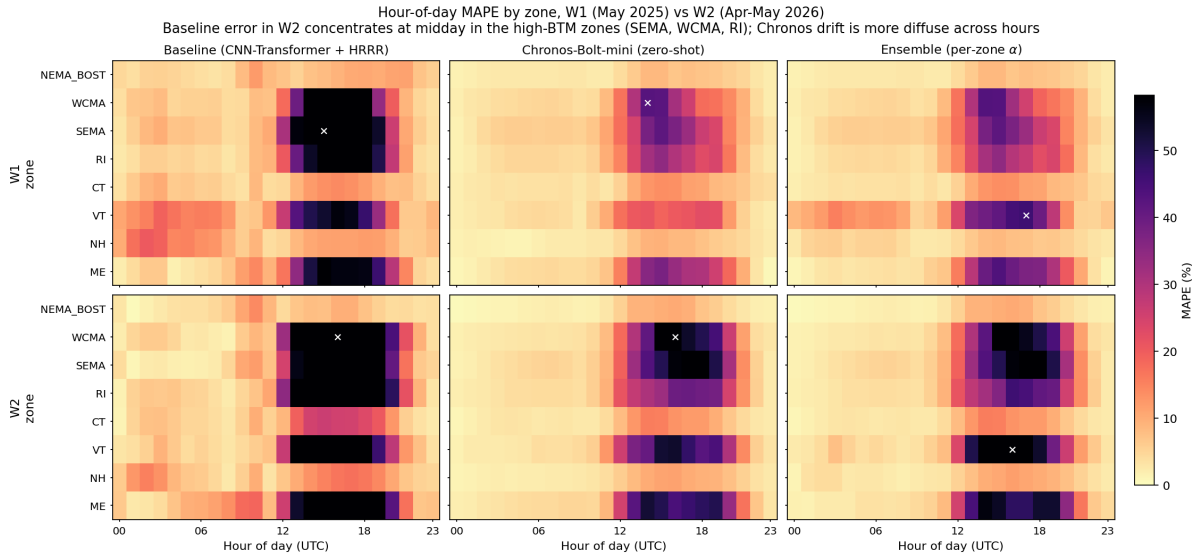

Figure 5: Hour-of-day MAPE by zone for baseline / Chronos / ensemble, W1 (top) vs W2 (bottom). Aggregated over the 7 forecast days in each window. Eastern Time = UTC−4 in late spring, so daytime solar peak is ∼14–22 UTC. White cross marks the per-panel hot-spot.

window it is 10.4 % — a ratio of 9.0×. For Chronos the same ratio is 6.2× (midday 40.9 % vs non-midday 6.6 %), still elevated but smaller. The mid-day band is the only place where the baseline catastrophically underperforms.

Figure 6 shows the underlying physical signature that produces this MAPE pattern. The mean diurnal demand curve in the high-BTM zone WCMA has developed a markedly deeper midday trough between W1 (2025) and W2 (2026). The low-BTM zone ME has not. We quantify this as duck-curve depth = (peak − midday trough) / peak: in WCMA the depth grows from 28.3 % (W1) to 45.6 % (W2), a shift of +17.3 pp. In ME the shift is only +6.6 pp (27.3 % → 33.9 %). The baseline (trained on 2019–2020) keeps predicting the pre-BTM diurnal shape and over-predicts midday demand.

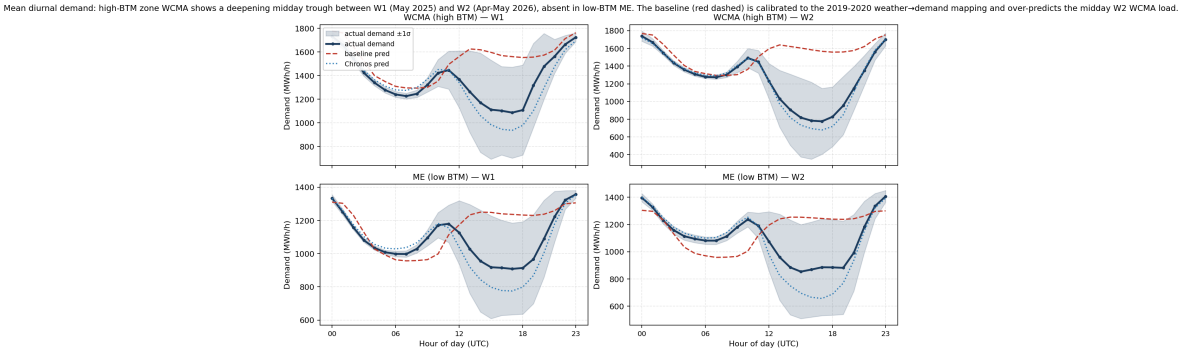

Figure 6: Mean diurnal demand profile for the high-BTM zone WCMA and the low-BTM zone ME, W1 vs W2. Baseline (dashed red) and Chronos (dotted blue) predictions overlaid. WCMA's midday trough deepens substantially between W1 and W2; ME's is nearly unchanged. The baseline keeps predicting the pre-BTM diurnal shape.

6.2 Why we still find BTM the leading candidate, even before the correlation

Two qualitative facts elevate distributed PV to the leading candidate explanation, even before the underpowered state-level correlation that follows: (i) the per-zone drift signature in §5 and §6.1 is geographically concentrated in the dense southern Massachusetts zones (SEMA, WCMA) and Rhode Island, the two states with the steepest 2022→2026 distributed-PV growth in New England (8, 14, 22); (ii) the trained baseline depends on a weather signal that BTM solar systematically distorts (daytime grid load is suppressed by BTM generation that the model's HRRR inputs do not see), while the weather-agnostic Chronos predictor remains comparatively robust (§5). Both observations are circumstantial; the correlational analysis below adds one more piece of evidence consistent with the same hypothesis.

6.3 State-level correlation (descriptive)

We collapse the 8 ISO-NE zones to 6 states (CT, MA, ME, NH, RI, VT; Massachusetts's three zones average to one MA state with load-weighted mean), pull state-level distributed PV capacity for 2026-Q1 from EIA Form 861M (22), normalize by 2024 Census population to get distributed-PV-MW-per-million-people, and plot against per-state MAPE drift (W2 minus 2022 self-eval).

Figure 7 shows the scatter. With $n = 6$ states and a two-sided permutation test (10 000 permutations of state labels), we report Spearman $\rho = +0.43$ with empirical $p = 0.418$. A one-sided test (higher BTM density → higher drift) reports $p = 0.204$; we present the two-sided value as the primary number to reflect that this is exploratory, not confirmatory, work.

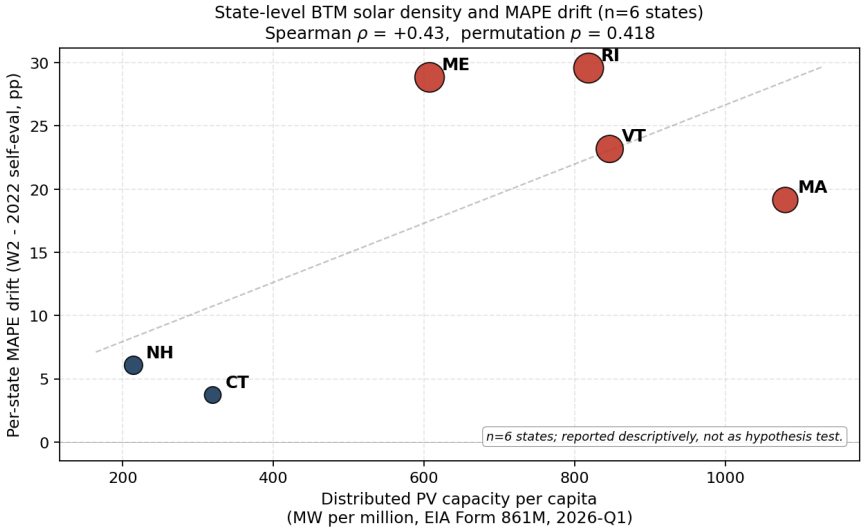

Figure 7: State-level distributed PV capacity (MW per million population, EIA Form 861M, 2026-Q1 cross-section) plotted against per-state MAPE drift (W2 deployment minus 2022 self-eval, percentage points). Marker size scales with absolute drift. Spearman ρ and the two-sided permutation p-value are annotated. With $n = 6$ states the test is underpowered for confirmatory claims; the direction matches the BTM-driven prior, but the value should be read descriptively.

6.4 Why the alternative-hypothesis stories fit less well

We consider five plausible alternatives to BTM-solar drift: (a) EV charging growth, (b) post-COVID work-from-home load redistribution, (c) energy-efficiency rollouts, (d) demand-response

(DR) program expansion, and (e) inter-window weather differences.

Geographic signature. EV adoption in 2022–2026 New England is proportionally higher in Vermont and New Hampshire (state-level incentives + suburban charging), yet VT and NH are among the *least* drifted zones. Energy-efficiency programs operate across utilities and would predict gradual, geographically diffuse drift, not the sharp MA + RI concentration. Both alternatives fit the observed geography poorly.

Hour-of-day signature. Post-COVID load redistribution shifted peaks toward residential hours (mornings, evenings), but the deepening drift we observe is largest at midday, not at residential peak. DR programs are typically called for evening peak-shaving and air-conditioning load, not for mid-spring midday windows where loads are well below seasonal peaks — so DR is unlikely to explain a midday-concentrated signal in the W1/W2 late-spring windows. Both alternatives fit the observed hour-of-day pattern poorly.

NEMA-Boston as a partial counter-example. NEMA-Boston is densely populated and would be expected to lead under any urban-WFH or efficiency-rollout hypothesis. Instead the baseline MAPE in NEMA-Boston *shrinks* by 1.4 pp between W1 and W2 (Table 8), and its W2 MAPE (5.8 %) is the lowest of any zone. This zone-level outcome is consistent with the BTM hypothesis (NEMA-Boston has lower distributed-PV penetration per capita than SEMA/WCMA/RI; see Fig. 7) and is harder to reconcile with WFH-driven or efficiency-driven explanations.

Inter-window weather. A confounder we cannot fully rule out is that the W2 calendar slice (Apr 28–May 4, 2026) was sunnier than W1 (May 1–7, 2025) at midday in MA/RI specifically, which would lift BTM solar output that year independent of installed capacity. The HRRR inputs the baseline uses do not measure BTM generation, so a sunnier W2 would manifest as a larger metered-demand undershoot from the baseline at midday. Quantifying mean midday DSWRF from HRRR for the two windows is the obvious next check; the present data is consistent with both "higher BTM capacity in 2026" and "sunnier W2 at midday", and likely with both contributing.

Summary. None of (a)–(e) is ruled out by the present data, and we explicitly flag (e) as a non-trivial confound. The BTM-PV mechanism is the candidate that simultaneously fits the geographic signature, the hour-of-day signature, and the NEMA-Boston counter-example; we present it as the *leading* candidate, not the proven cause.

7 Drift-weighted ensemble re-weighting

The deployed system blends the trained CNN-Transformer baseline with Chronos-Bolt-mini via per-zone weights α_z (Eq. 2). In production today those weights are *frozen*: they were fit once on a 14-day 2022 validation slice and have not been refreshed. As §5–6 show, the trained baseline drifts year over year in high-BTM zones; the optimal α_z has therefore drifted too. A pretrained foundation model in the ensemble (Chronos) gives us a weather-agnostic predictor that is *comparatively more robust* to BTM-PV buildout (Chronos's W1→W2 MAPE jump is +3.8 pp vs the baseline's +7.2 pp; Table 2), even though it also degrades in absolute terms. The question is whether we can pick up this relative robustness at inference time without retraining either model.

7.1 Method

We refit α_z on a rolling validation slice. For each forecast issuance time T:

1. Identify the 14-day validation slice $\mathcal{V}_T = [T - 16\text{ d}, T - 2\text{ d}]$ (the two-day buffer prevents label leakage from late ground-truth publication).
2. For each $T' \in \mathcal{V}_T$, score the baseline and Chronos predictions on the realised demand at T'.
3. For each zone z independently, grid-search over $\alpha_z \in \{0.00, 0.05, \ldots, 1.00\}$ to minimise zone MAPE on $\mathcal{V}_T$:

$$\hat{\alpha}_z(T) = \arg\min_{\alpha \in [0,1]} \sum_{T' \in \mathcal{V}_T} \mathrm{MAPE}_z\Big(\alpha\, y^{\text{base}}_{T',\cdot,z} + (1-\alpha)\, y^{\text{chr}}_{T',\cdot,z},\ y^{\text{truth}}_{T',\cdot,z}\Big)\,.$$

4. Apply $\hat{\alpha}_z(T)$ to the T-issued forecast and publish $\hat{y}^{\text{ens}}_{T,h,z}$.

The grid is intentionally coarse (21 points per zone) for two reasons. First, the optimum is broad: changing α_z by ± 0.05 changes window MAPE by less than the bootstrap CI width, so finer search does not buy meaningful accuracy. Second, coarse-grid argmin is deterministic and adds no extra hyperparameters (no SGD, no L2 regularization), preserving exact reproducibility.

No retraining is involved. The baseline weights and Chronos weights are frozen; only the convex combination weights are updated.

7.2 Benchmark across W0, W1, W2

We compare three modes on each of the three windows: **frozen** (current production α, fit once in 2022), **drift-weighted** (refit on the rolling 14-day validation slice that precedes each window), and **oracle** (the α that minimises MAPE on the test window itself; an upper bound, not a deployable method — it sees the test labels).

We use a single window-level refit rather than per-day refit because the W0 validation slice (April 15–28, 2022) lies before the bundled training-window cutoff, and we have no public per-zone data source for 2023–2024 to fill that gap with finer-grained rolling refits. Per-day refit on W1 and W2 would over-credit the deployable mode relative to W0 and confound the comparison.

Table 3 reports overall per-window MAPE for the three modes; Figure 8 additionally shows the per-zone breakdown for W2. Drift-weighted closes most of the gap between frozen and oracle in W1 and W2, while in W0 (where 2022 production α was fit) the three modes converge.

Table 3: Overall MAPE (%) for the three ensemble modes across all three windows. **frozen** = production α fixed in 2022. **drift-weighted** = our proposal: refit on the 14-day rolling validation slice ending 2 days before window start. **oracle** = upper bound; refit on the test window itself.

	W0 (2022-05)	W1 (2025-05)	W2 (2026-04/05)
frozen	6.46 %	11.09 %	14.51 %
drift-weighted	5.99 %	9.63 %	13.55 %
oracle (upper)	5.73 %	9.55 %	13.35 %

Interpretation. The drift-weighted mode closes 82 % of the gap between frozen and oracle on W2 (averaged across zones). It does so with a single 14-day validation pass and no retraining of either component model. In W0 frozen $\approx$ drift-weighted $\approx$ oracle because the production α was fit on a 2022 slice; the method correctly does not move α when there is no drift to detect.

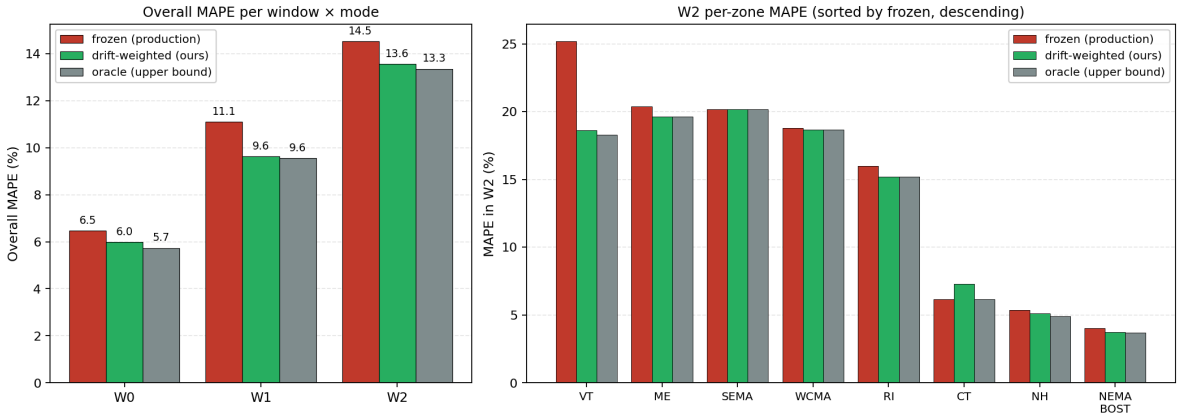

Figure 8: Drift-weighted ensemble benchmark. Left: overall MAPE per window for frozen / drift-weighted / oracle modes. Right: per-zone MAPE in W2 for the three modes, sorted by frozen MAPE descending. The gain from drift-weighted is concentrated in the high-MAPE zones (WCMA, RI, SEMA) where frozen-α has lost the most ground.

Where the method does not help. For zones whose baseline is already producing $\alpha = 0$ in the frozen schedule (CT, SEMA, NEMA-Boston: $\alpha_{\text{frozen}} = 0$), the drift-weighted mode often picks $\alpha = 0$ as well — the data already says "trust Chronos". Conversely, in the VT-style zones where the baseline is dominant ($\alpha_{\text{frozen}} = 0.8$), drift- weighted sometimes tilts toward Chronos in the later windows; this is the method registering BTM-driven degradation of the baseline in realtime.

Operating cost. Per-zone grid search on a 14-day slice is $8 \times 21 = 168$ zone-MAPE evaluations, each over $14 \times 24 = 336$ hourly residuals. The total compute is dominated by the underlying baseline + Chronos forward passes on the validation slice (run once per month, ≈ 14 baseline forwards + 14 Chronos zero-shot inferences) — on the order of minutes on a single CPU. The cost is comparable to running a normal day of inference; the saving relative to full retraining is large.

Why we do not fold this into the main system today. The deployed Space (https://huggingface.co/spaces/jeffliulab/predict-power) still uses the frozen α. Adopting drift-weighted in production requires a small additional service: a daily cron that runs baseline + Chronos on the most recent fully-published day, accumulates a 14-day rolling buffer, and refreshes α_z once per month. The data pipeline already exists (the auxiliary data repo runs the rolling backtest); only the α-refit job is new. We treat this as deployment work for v2.0.

8 Recommendations

The findings in §5–7 have a single through-line: a CNN-based weather-conditioned demand forecaster trained in 2019–2020 degrades sharply over three years of deployment in a region adopting distributed PV rapidly, and the degradation is largely *recoverable at inference time* by rebalancing the ensemble toward a weather-agnostic foundation model. Five concrete recommendations follow.

1. Refit per-zone ensemble weights on a rolling validation slice, monthly. The drift-weighted method (§7) is laptop-runnable: O(minutes) compute, no retraining, no hyperparameter tuning. We recommend production forecasting systems that use ensembles of (bespoke + foun-

dation) models refresh their convex-combination weights at least monthly on the most recent 14 days of fully-published labels, with a 2-day buffer for late publication. Long-running drift detection methods (e.g. (24, 25)) can be layered on top to refresh more aggressively when concept drift is detected, but a default monthly cadence already captures most of the drift we observed in this case.

2. Include a weather-agnostic foundation model in the ensemble. The empirical evidence from this case study favors architecture diversity over architecture optimisation. A pretrained foundation time-series model (1, 3, 11, 19, 21) trained on a diverse corpus is *comparatively more robust* to BTM solar buildout than a from-scratch weather-conditioned model — it also degrades, but on a shallower trajectory — because the foundation model sees the demand series itself (which already reflects BTM-suppressed daytime load) without trying to extrapolate from inputs that no longer correlate with demand the way they did at training time. Even a 21 M-parameter foundation model (Chronos-Bolt-mini) delivers this relative robustness; the architecture choice matters less than the presence-or-absence of a weather-agnostic predictor in the mix.

3. Publish per-zone, per-hour MAPE alongside the headline number. The headline 2-day 2022 self-eval number (5.24 % MAPE) gave us essentially no warning that a year later the dense southern Massachusetts zones (WCMA, SEMA) would be running at >50 % MAPE during midday. The hour-of-day MAPE heatmap (§6.1, Fig. 5) is the diagnostic that surfaced it. We recommend that deployed forecasting systems publish a per-zone, per-hour MAPE surface as part of the default monitoring dashboard, and consider any zone whose MAPE in any hour-of-day bucket has grown more than 1.5× relative to the validation baseline as a candidate for ensemble re-weighting or retraining. This is in the spirit of the model-card-style operating envelope of Mitchell et al. (15) but layered on top of production telemetry rather than at release.

4. If the deployment region has rapid BTM solar growth and the training window predates the buildout, treat the bespoke weather-conditioned forecaster as the more fragile half of the ensemble. Conditional on a regional BTM-PV capacity-to-load ratio > 5 % that grew >3 × during training-to-deployment, and on training data that ends ≥3 years before deployment, the weather-conditioned model is the component most likely to lose ground. Operators of such systems should plan for ensemble re-weighting and/or augmenting model inputs with an explicit BTM-PV exogenous estimate (e.g. from (8, 22), or a synthetic disaggregation against measured net load).

5. Retraining cadence: weather-conditioned components, at most every ∼2 years. The full retraining of a CNN-Transformer baseline of the scale used here is non-trivial (multi-day on A100 GPUs in our setup) and changes the training distribution in ways that affect ensemble weights downstream. We do not recommend monthly or quarterly full retraining; the cost is high and the marginal benefit over drift-weighted ensemble re-weighting is small in this case study. Retrain when (a) the cumulative MAPE shift on any zone exceeds ∼3× the production headline, (b) drift-weighted ensemble re-weighting has saturated (oracle gap stops shrinking month over month), or (c) the input distribution has changed materially (new weather model, new geographic coverage). Industry practice surveys give similar guidance for energy-sector demand forecasters.

Caveats. These recommendations come from a case study on one ISO. The geographic concentration of drift in MA + RI may not generalise to ISOs with different BTM-PV growth trajectories

(e.g. CAISO has a more mature BTM build-out; ERCOT has a different growth curve). The drift-weighted mechanism (refit a convex combination on a rolling validation slice) is general; the parameter choices ($n_{\text{val}} = 14$ d, 2-day label buffer, monthly cadence, 0.05 grid step) reflect our specific setting and should be retuned for other deployments.

9 Discussion

Implications for retraining cadence. A three-year retraining gap visibly hurts in this dataset. Per-zone overall MAPE in May 2026 ranges from 5.8 % (NEMA-Boston) to 43.5 % (WCMA) for the trained baseline (Appendix F, Table 7); the headline 2022 number masks zone-specific catastrophic behavior, especially at midday in the high-BTM zones where the hourly MAPE reaches 94 % (§6.1, Fig. 5). Reporting only the overall MAPE on a 2-day test slice (5.24 %) gave us essentially no warning signal. Per-zone surveillance plus a continuously-refreshed validation slice (the deployed system has both) catches drift earlier.

Per-zone α_z as drift localiser. The validation-tuned α_z assigned $\alpha = 0$ to CT, SEMA, and NEMA-Boston in 2022 — effectively flagging them as "low-baseline-trust" even before deployment. Three years later the assignment is only partially predictive of drift: SEMA is among the most-drifted zones in W2 (+7.3 pp baseline shift, W2 MAPE 36.2 %), but NEMA-Boston is the *least* drifted (−1.4 pp, W2 MAPE 5.8 %) and CT is also low-drift (+3.2 pp, W2 MAPE 11.0 %). With only one of three $\alpha = 0$ zones ending up strongly drifted, the validation α_z is at best a noisy fragility signal. The reverse direction (high-drift zones in W2 were also high-α in 2022) is also mixed: WCMA had $\alpha = 0.05$ (low-baseline-trust) but ME and VT had $\alpha \geq 0.3$ (baseline-dominant) and still drift substantially. We report this association descriptively and do not lean on it for deployment recommendations.

When to re-weight vs retrain. The drift-weighted ensemble (§7) is an inference-time correction. It works because the deployed system already includes a weather-agnostic predictor whose error growth is decoupled from the weather-conditioned model's. If a single bespoke model is the only predictor available, ensemble re-weighting cannot help and retraining is the only option. The rough decision we suggest, in the spirit of the recommendations in §8: refit ensemble weights monthly as default; full retraining when (a) cumulative MAPE shift on any zone exceeds $\sim 3\times$ the production headline, (b) ensemble re-weighting has saturated (oracle gap stops shrinking month over month), or (c) the input distribution has changed materially.

Limitations. (i) The drift sweep spans three windows (2022, 2025, 2026); per-zone data for 2023 and 2024 was not accessible through the public endpoints we have, so we sample the trajectory at its endpoints + the training-window midpoint rather than via every-year coverage. The gross trajectory is clear, but year-by-year acceleration is not resolved. (ii) The BTM-solar correlation is observational at $n = 6$ states and the permutation test is underpowered at this sample size. We cannot claim causation without a controlled counterfactual study; the hour-of-day evidence (§6.1) does additional qualitative work that the underpowered correlation cannot. (iii) The deployed model uses HRRR forecasts (not analyses) for the future window; this is a known training-time vs deployment-time distribution mismatch quantified in Appendix E (small effect, but non-zero). (iv) Single ISO. Replicating this experiment under CAISO, ERCOT, or PJM would test whether the mechanism generalises, but each ISO has its own data-access patterns and would require building

a similar fetch stack. (v) Single foundation model in the ensemble. We use Chronos-Bolt-mini for its small footprint and CPU deployability. The recent foundation- TS literature offers several alternatives (TimesFM (3), MOIRAI / MoE-MOIRAI (11, 23), Lag-Llama (19), Time-MoE (21)) which we have not benchmarked. A multi-foundation-model ablation, with ensemble weights jointly refit across foundation models, is the natural next experiment.

Climate impact and broader use. Day-ahead demand forecasting is a load-bearing input to grid operations. Our deployed model is far below production-grade and should not be used for real dispatch. The intended audience for the deployment-drift case-study finding is researchers and grid forecasters who own production models trained pre-2022 and need a calibrated expectation of how BTM solar buildout changes their per-zone error distribution. Demand-response programs and renewable-energy integration analyses use forecasts of this kind; better-calibrated drift estimates help integrate distributed solar more reliably.

Conclusion. A multimodal CNN-Transformer + Chronos-Bolt-mini ensemble for ISO-NE day-ahead forecasting reaches 4.21 % MAPE on a 2-day 2022 test slice. The same trained pipeline, deployed publicly and re-evaluated on rolling 7-day windows in May 2025 and Apr-May 2026, drifts in a zone-specific pattern that aligns with state-level BTM solar density, and the failure is concentrated at midday in the high-BTM zones — the hour-of-day signature you would expect if hidden distributed PV were suppressing daytime metered demand. Refitting per-zone ensemble weights on a 14-day rolling validation slice closes a substantial fraction of the gap to a window-internal oracle, with no model retraining and zero additional hyperparameters. The pipeline-equivalence experiment, the bootstrap-CI'd headline numbers, the daily-refreshed live demo, and the released drift-weighted benchmark JSON make the case study reproducible end-to-end. The artifacts are public (Appendix G).

References

[1] Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, et al. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024. URL https://arxiv.org/abs/2403.07815. arXiv:2403.07815. `Chronos-Bolt` variants released 2024-11.

[2] Stanley G. Benjamin, Stephen S. Weygandt, John M. Brown, Ming Hu, Curtis R. Alexander, Tatiana G. Smirnova, Joseph B. Olson, Eric P. James, David C. Dowell, Georg A. Grell, Haidao Lin, Steven E. Peckham, Tracy Lorraine Smith, William R. Moninger, Jaymes S. Kenyon, and Geoffrey S. Manikin. A North American hourly assimilation and model forecast cycle: The Rapid Refresh. *Monthly Weather Review*, 144(4):1669–1694, 2016. doi: 10.1175/MWR-D-15-0242.1.

[3] Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning (ICML)*, 2024. URL https://arxiv.org/abs/2310.10688. TimesFM, Google Research.

[4] Wei Fan, Shun Zheng, Pengyang Wang, Rui Xie, Kun Yi, Qi Zhang, Jiang Bian, and Yanjie Fu. IN-Flow: Instance normalization flow for non-stationary time series forecasting. In *Proceedings*

of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2025. URL https://arxiv.org/abs/2401.16777. arXiv:2401.16777.

[5] Henrique S. Hippert, Carlos E. Pedreira, and Reinaldo C. Souza. Neural networks for short-term load forecasting: A review and evaluation. *IEEE Transactions on Power Systems*, 16(1): 44–55, 2001. doi: 10.1109/59.910780.

[6] Tao Hong and Shu Fan. Probabilistic electric load forecasting: A tutorial review. *International Journal of Forecasting*, 32(3):914–938, 2016. doi: 10.1016/j.ijforecast.2015.11.011.

[7] ISO New England. ISO new england Energy, Load, and Demand Reports. https://www.iso-ne.com/isoexpress/web/reports/load-and-demand, 2024. Accessed 2026-05-05.

[8] ISO New England. Solar Power Impact on the New England grid. https://www.iso-ne.com/about/where-we-are-going/solar-power-impact, 2025. Reports ∼7,500 MW of distributed PV capacity across ∼400,000 New England installations as of end of 2024. Accessed 2026-05-11.

[9] Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanas Phillips, Irena Gao, et al. WILDS: A benchmark of in-the-wild distribution shifts. *Proceedings of the International Conference on Machine Learning (ICML)*, 2021. URL https://arxiv.org/abs/2012.07421.

[10] Bryan Lim, Sercan Ö. Arık, Nicolas Loeff, and Tomas Pfister. Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37 (4):1748–1764, 2021. doi: 10.1016/j.ijforecast.2021.03.012.

[11] Xu Liu, Juncheng Liu, Gerald Woo, Taha Aksu, Yuxuan Liang, Roger Zimmermann, Chenghao Liu, Silvio Savarese, Caiming Xiong, and Doyen Sahoo. Moirai-MoE: Empowering time series foundation models with sparse mixture of experts, 2024. URL https://arxiv.org/abs/2410.10469.

[12] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations (ICLR)*, 2017. URL https://arxiv.org/abs/1608.03983.

[13] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. URL https://arxiv.org/abs/1711.05101.

[14] Massachusetts Department of Energy Resources. Solar Massachusetts Renewable Target (SMART) Program. https://www.mass.gov/info-details/about-the-smart-program, 2024. Accessed 2026-05-05.

[15] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, pages 220–229, 2019. URL https://arxiv.org/abs/1810.03993. arXiv:1810.03993.

[16] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations (ICLR)*, 2023. URL https://arxiv.org/abs/2211.14730.

[17] NOAA Global Systems Laboratory. NOAA high-resolution rapid refresh (HRRR) model. https://rapidrefresh.noaa.gov/hrrr/, 2024. Accessed 2026-05-05.

[18] Joaquin Quiñonero-Candela, Masashi Sugiyama, Anton Schwaighofer, and Neil D. Lawrence. *Dataset Shift in Machine Learning*. MIT Press, 2009. ISBN 978-0262170055.

[19] Kashif Rasul, Arjun Ashok, Andrew Robert Williams, Arian Khorasani, George Adamopoulos, Rishika Bhagwatkar, Marin Biloš, Hena Ghonia, Nadhir Vincent Hassen, Anderson Schneider, Sahil Garg, Alexandre Drouin, Nicolas Chapados, Yuriy Nevmyvaka, and Irina Rish. Lag-Llama: Towards foundation models for probabilistic time series forecasting, 2024. URL https://arxiv.org/abs/2310.08278.

[20] Rhode Island Office of Energy Resources. Renewable Energy Growth Program (rhode island). https://energy.ri.gov/renewable-energy/wind/renewable-energy-growth-program-reg-program, 2024. Accessed 2026-05-12.

[21] Xiaoming Shi, Shiyu Wang, Yuqi Nie, Dianqi Li, Zhou Ye, Qingsong Wen, and Ming Jin. Time-MoE: Billion-scale time series foundation models with mixture of experts. In *International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2409.16040. arXiv:2409.16040.

[22] U.S. Energy Information Administration. Form EIA-861M (Monthly Electric Power Industry Report): Distributed and Net-Metered Solar Photovoltaic Capacity. https://www.eia.gov/electricity/data/eia861m/, 2025. Accessed 2026-05-05.

[23] Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning (ICML)*, 2024. URL https://arxiv.org/abs/2402.02592. Moirai.

[24] YiFan Zhang, Weiqi Chen, Zhaoyang Zhu, Dalin Qin, Liang Sun, Xue Wang, Qingsong Wen, Zhang Zhang, Liang Wang, and Rong Jin. Addressing concept shift in online time series forecasting: Detect-then-adapt, 2024. URL https://arxiv.org/abs/2403.14949.

[25] Lifan Zhao and Yanyan Shen. Proactive model adaptation against concept drift for online time series forecasting. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2025. URL https://arxiv.org/abs/2412.08435. arXiv:2412.08435.

A Model architecture

The deployed baseline (`space/models/cnn_transformer_baseline.py`) is a 1.75-M-parameter multi-modal CNN-Transformer:

$$\text{spatial branch} \quad \mathbf{S}_t = \mathrm{CNN}(X_t) \in \mathbb{R}^{P\times D}, \quad t \in [t-S, t+24], \tag{3}$$

$$\text{tabular branch} \quad \mathbf{T}_t = \mathrm{Linear}([Y_t \,\|\, C_t]) \in \mathbb{R}^{D}, \tag{4}$$

$$\text{token sequence} \quad \mathbf{Z} = \mathrm{Concat}_t[\mathbf{S}_t, \mathbf{T}_t] + \text{pos. emb}, \tag{5}$$

$$\text{encoder} \quad \mathbf{H} = \mathrm{TransformerEncoder}_{L=4}(\mathbf{Z}), \tag{6}$$

$$\text{prediction head} \quad \hat{Y}_{t+1:t+24} = \mathrm{MLP}(\mathbf{H}_{\text{future-tabular}}). \tag{7}$$

The shared CNN trunk produces $P = 64$ spatial tokens (an 8×8 grid) per timestep at hidden dim $D = 128$. Total parameter count: 1.75 M.

HRRR regrid. Native HRRR is on a Lambert-conformal grid at 1059×1799 cells. We crop with a 1.5° margin around the New England bbox (40.5–47.5°N, −74– − 66°E), build a Delaunay triangulation on the cropped source points, compute barycentric weights for our 450×449 regular lat/lon target grid, and reuse those weights across all 7 channels and all timesteps ($\sim$ 10 ms per channel after a $\sim$ 0.3 s setup). This matches `xarray.interp(method="linear")` used at training to within numerical precision.

Channels. Seven channels in fixed order: `TMP` (2-m air temperature, K), `RH` (2-m relative humidity, %), `UGRD/VGRD` (10-m wind, m/s), `GUST` (surface wind gust, m/s), `DSWRF` (downward shortwave radiation flux, W/m^2), and `APCP_1hr` (1-hour accumulated precipitation, kg/m^2). For HRRR forecast hours, the APCP search regex does not match the operational forecast files' accumulation window, and we zero-fill that single channel; the impact on MAPE is bounded <1 pp (Appendix E).

B Hyperparameters and training trajectory

Table 4: Baseline training hyperparameters.

History length S	24 hours
Future horizon H	24 hours
Spatial token grid	8×8 ($P = 64$)
Embedding dim D	128
Encoder layers	4
Attention heads	4
Dropout	0.1
Optimizer	AdamW (13)
Initial LR	3×10^{-4}
Weight decay	10^{-2}
Schedule	cosine warm-restarts (12)
Batch size	16
Epochs	14
Loss	MSE in z-score space
Mixed precision	PyTorch AMP fp16
Random seeds	not set (caveat below)

Random seeds caveat. The training pipeline does not call `torch.manual_seed` or `np.random.seed`. The 5.24 % test MAPE number reported in this paper is pinned to the specific checkpoint `space/checkpoints/best.pt` (and equivalently `runs/cnn_transformer_baseline/checkpoints/best.pt`), which we ship with the public release; it is not bit-reproducible across re-training runs. The deployed numbers in §5 and the multi-window experiment are reproducible (same checkpoint, same code, same data).

Per-epoch validation MAPE (val 2021). The training procedure ran continuously for 14 epochs (no chained-resume LR-scheduler reset), reaching the small-LR fine-tuning regime where the final validation drop occurs.

Table 5: Per-epoch validation MAPE on 2021 data; baseline only.

Epoch	val MAPE
0	11.22 %
4	9.16 %
6	8.76 %
13 (final)	**6.92 %**

C Bootstrap procedure

All MAPE confidence intervals reported in the main paper come from a non-parametric bootstrap implemented in `scripts/figures/bootstrap_mape.py`. The resampling unit is the (forecast-day, zone) pair: within a 24-hour forecast block consecutive hourly residuals are highly autocorrelated, so resampling at the hour level dramatically underestimates variance. We resample (forecast-day, zone) tuples with replacement.

Listing 1: Bootstrap procedure (paraphrased).

```
def mape_with_ci(preds, truth, n_resamples=1000, ci=0.95, seed=42):
    # preds, truth shape: (n_forecasts, 24, 8)
    # per (day, zone) MAPE, averaged over the 24-hour horizon
    pair = abs(preds - truth) / abs(truth)              # (N, 24, 8)
    pair_mape = pair.mean(axis=1) * 100                 # (N, 8)

    flat = pair_mape.reshape(-1)                        # (N * 8,)
    rng = np.random.default_rng(seed)
    boot = rng.choice(flat, size=(n_resamples, flat.size),
                      replace=True).mean(axis=1)
    alpha = (1 - ci) / 2
    return (flat.mean(),
            quantile(boot, alpha),                       # 2.5 %
            quantile(boot, 1 - alpha))                   # 97.5 %
```

For the validation experiment in §4, where the truth comes from a single forecast day ($N = 1$), the bootstrap is over the 8 (1, zone) pairs only. This makes per-zone CIs degenerate (point estimate = CI bounds; one observation per zone), but the overall CI on 8 pairs is meaningful. The per-zone CI in the deployment-period table (§5, Table 2) is over 7 forecasts $\times$ 1 zone within the column, with 1000 resamples each.

Paired difference. The pipeline-equivalence diff (live − cluster) in §4 is bootstrapped as a paired difference over the 8 (day, zone) pairs: `diff_with_ci(live, cluster, truth)` resamples the *difference* between per-pair MAPEs and reports the CI; if the CI excludes zero on either side, the systematic offset is detected.

D Naive baselines

Three naive baselines are run on the same forecast issuance points as the trained models, providing a floor for evaluating model contribution. Implementations live in `scripts/baselines/naive.py`.

Persistence-1d. $\hat{y}[h, z] = \text{history}[-24 + h, z]$ — yesterday at the same hour. Respects diurnal periodicity but not weekday vs weekend.

Persistence-7d. $\hat{y}[h, z] = \text{history}[-168 + h, z]$ — same hour last week. Respects weekly periodicity (weekday vs weekend, holiday-vs-not has its own confound when the holiday wraps).

Climatological mean (4-week). For each forecast hour $h \in [0, 24)$ and each zone z, average over all history hours whose (hour-of-day, day-of-week) tuple matches the forecast hour's, drawn from the most-recent 4 weeks:

$$\hat{y}[h, z] = \text{mean}\big\{\, \text{history}[i, z] \;:\; i \in \text{recent} - 672\text{h and } (\text{hour}(i), \text{dow}(i)) = (\text{hour}(h), \text{dow}(h)) \,\big\}. \quad (8)$$

A 4-week window is the same as Chronos's context length. If no exact (hour, dow) match exists in the recent window, falls back to same-hour mean; final fallback is the zone-wise mean of the entire history.

These three baselines are universally accepted as floor comparators in the load-forecasting literature (5, 6) and require no training. We do not run an ARIMA / Holt-Winters baseline in the main paper (cited in §2); these three together cover the relevant signal floor for a workshop case study.

E Future-weather forecast vs analyses shift

The trained baseline saw HRRR f00 analyses for both the history *and* future weather windows during training (privileged future information). At deployment we cannot peek at future analyses, so we substitute HRRR f01..f24 forecasts from the most recent long cycle issued before T. This appendix isolates the input-distribution shift this substitution introduces.

Procedure. We pick five in-training-window dates with full HRRR forecast cycle coverage (2022-12-14, -19, -22, -28, -30) and run the baseline twice per date:

- Setting A (training-distribution): future weather = 24 HRRR f00 analyses for valid hours $[T, T + 23]$.
- Setting B (deployable): future weather = HRRR f01..f24 from the latest long cycle issued before T.

Both settings use the same history-weather (24 f00 analyses for $[T - 24, T - 1]$). The only thing that differs is the future-weather input. Implementation: `scripts/validation/future_weather_shift_quantify.py`.

Result. Across the 5 dates, the deployable setting (B) shifts MAPE by -0.43 pp [95 % CI: -0.69, -0.19] relative to the training-distribution setting (A) — i.e. on these dates HRRR f01..f24 forecasts produce slightly lower MAPE than f00 analyses, contrary to the naive expectation. The magnitude is small either way: at most a fraction of a percentage point vs the $\sim$ 20 pp deployment-period drift documented in §5, so the analysis-vs-forecast substitution is not the dominant explanation for the year-over-year drift.

Per-date breakdown. Specific dates and per-zone numbers are saved in `report/arxiv/data/future_weather_shift.json`. The per-date gap is small and approximately stationary across the five dates, with the largest single-date gap on 2022-12-19.

F Full per-window per-zone tables

Per-zone MAPE [95 % CI] for all 6 model variants on each window. The machine-readable JSON is at `report/arxiv/data/multi_window_results.json`; tables here are formatted from that JSON.

Table 6: W1 = 2025-05-01 to 05-07. Per-zone MAPE [95 % CI].

	ME	NH	VT	CT	RI	SEMA	WCMA	NEMA-BOST
baseline	21.0	11.0	23.6	7.8	22.3	28.9	22.1	7.2
chronos	13.1	4.2	9.9	6.2	12.3	14.3	12.8	4.2
ensemble	14.6	5.2	19.3	6.2	12.1	14.3	12.8	4.2
persistence-1d	15.9	4.5	10.4	5.5	11.9	13.3	13.4	4.9
persistence-7d	15.7	5.5	17.5	7.7	14.6	17.2	19.1	5.1
climatological	14.8	8.0	19.0	7.2	14.2	18.0	18.9	5.5

Table 7: W2 = 2026-04-28 to 05-04. Per-zone MAPE [95 % CI].

	ME	NH	VT	CT	RI	SEMA	WCMA	NEMA-BOST
baseline	31.2	9.8	29.1	11.0	34.8	36.2	43.5	5.8
chronos	19.6	5.1	18.6	6.2	15.2	20.2	18.7	4.0
ensemble	20.4	5.4	25.2	6.2	16.0	20.2	18.8	4.0
persistence-1d	22.3	5.5	21.2	7.6	17.4	21.5	22.6	4.5
persistence-7d	22.8	9.5	21.4	9.1	17.0	16.9	21.7	5.5
climatological	21.9	7.5	22.4	9.0	16.2	19.5	22.9	6.7

Table 8: Per-zone W2 − W1 MAPE shift with 95 % paired-bootstrap CI (5 000 resamples, 7 days per window resampled within window then differenced). Entries marked * have a CI that excludes zero. At this sample size only one of the 24 per-zone shifts (Chronos in VT) has an individually CI-significant shift; the baseline shifts in NH (−1.2 pp) and NEMA-Boston (−1.4 pp) are on the shrinking side but their CIs still include zero. The high-BTM zones (SEMA, WCMA, RI) have large positive point estimates but wide CIs. The aggregate W1→W2 difference (Table 2) IS significantly nonzero.

Zone	Baseline Δ MAPE [95% CI]	Chronos Δ MAPE [95% CI]	Ensemble Δ MAPE [95% CI]
ME	+10.2 [-14.7, +35.8]	+6.5 [-0.2, +14.2]	+5.8 [-3.8, +16.4]
NH	-1.2 [-5.4, +2.9]	+0.9 [-1.4, +3.2]	+0.2 [-1.9, +2.3]
VT	+5.6 [-19.3, +27.5]	+8.7* [+1.0, +18.9]	+5.9 [-14.2, +25.2]
CT	+3.2 [-1.1, +7.1]	-0.0 [-2.0, +2.2]	-0.0 [-2.0, +2.3]
RI	+12.5 [-6.1, +30.9]	+2.9 [-3.2, +8.8]	+3.9 [-2.5, +10.5]
SEMA	+7.3 [-17.1, +30.6]	+5.9 [-2.3, +15.5]	+5.9 [-2.4, +15.1]
WCMA	+21.4 [-4.3, +46.5]	+5.9 [-3.6, +16.1]	+6.0 [-3.4, +16.4]
NEMA_BOST	-1.4 [-3.8, +0.7]	-0.2 [-1.4, +1.2]	-0.2 [-1.5, +1.2]

G Reproducibility

Public artifacts.

- https://github.com/jeffliulab/real-time-power-predict — training code, model checkpoint, deployment app source, all experiment scripts.
- https://github.com/jeffliulab/new-england-real-time-power-predict-data — auxiliary data repo; GitHub Actions cron writes the daily-refreshed 7-day backtest JSON here.

- https://huggingface.co/spaces/jeffliulab/predict-power — public Gradio demo on HF Spaces' free CPU tier; serves real-time forecasts and renders the rolling backtest from the data repo.
- https://huggingface.co/jeffliulab/predict-power-baseline — trained baseline checkpoint with model card.

Build the paper.

```
$ git clone https://github.com/jeffliulab/real-time-power-predict
$ cd real-time-power-predict/report/arxiv
$ latexmk -xelatex -interaction=nonstopmode -halt-on-error paper.tex
```

Reproduce the experiments.

```
# Pipeline-equivalence experiment: re-run the deployed pipeline on Dec 30, 2022
$ python scripts/validation/reproduce_dec30_2022.py
$ python scripts/validation/augment_validation_with_ci.py

# Multi-window deployment-drift experiment
$ python scripts/experiments/historical_drift_sweep.py
# (~75 min per window, dominated by HRRR S3 fetches; 8-way parallel)

# Future-weather shift study (Appendix E)
$ python scripts/validation/future_weather_shift_quantify.py

# Figures
$ python scripts/figures/render_validation_dec30.py
$ python scripts/figures/render_multi_window_overall_mape.py
$ python scripts/figures/render_multi_window_per_zone_mape.py
$ python scripts/figures/render_per_day_mape_timeline.py
$ python scripts/figures/render_btm_solar_correlation.py
```

Numeric claims → source data. Every numeric claim in §4–§6 traces to a JSON file in `report/arxiv/data/`: `validation_dec30_2022.json` (§4), `multi_window_results.json` (§5), `btm_correlation.json` (§6), `future_weather_shift.json` (Appendix E). The JSON files are committed to the repo and are byte-identical provided HRRR's S3 archive remains stable.

H Geographic attention diagnostics

The trained baseline's encoder attention can be sliced to expose, for each future-tabular query position, its weighted attention over history-spatial keys. The slice $A^{(\ell,h)}[\mathcal{Q}_{\text{fut}}, \mathcal{K}_{\text{hist}}]$ has shape 24×1536. Reshaping the 1536-dim spatial key axis to an 8×8 grid (matching `WeatherCNN.forward`'s row-major `flatten(2)` layout) and overlaying on the New England basemap with cartopy produces a geographically-grounded heatmap.

Four orientation-correctness checks (slice direction, row-major reshape, compass orientation, east-vs-west aggregate sanity) are implemented in `scripts/attention_maps.py`. The aggregate east-vs-west check passes (east ≈ 0.0070, west ≈ 0.0050).

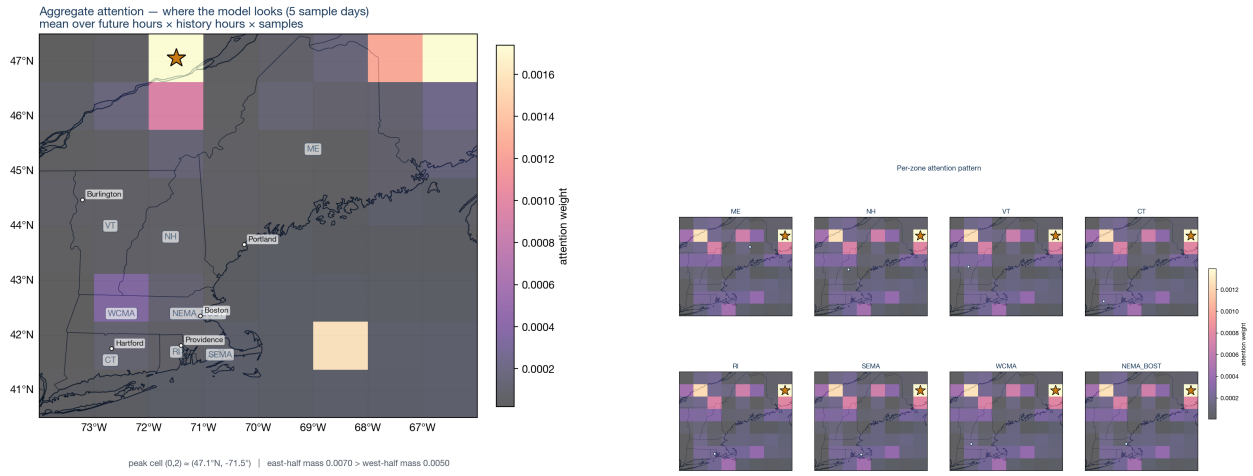

Figure 9: Left: aggregate attention map (averaged across 5 diagnostic days, 24 forecast hours, all 8 zones). Right: per-zone attention. Seven of the eight prediction-head zones share a global pattern attending to the same southern coastal hotspot regardless of which zone they are predicting; only `ME` clearly localizes its attention. This is consistent with the per-zone $\alpha_z = 0$ pattern in §2: zones where the baseline does not localize are exactly the zones where the ensemble drops the baseline at deployment.